

3. 傅里叶变换

一、周期信号的傅里叶级数分析

1. 三角函数形式的傅里叶级数

1) 三角函数集

$$\begin{aligned} & \langle f(t), f_1(t) \rangle = \int_{-\infty}^{\infty} f(t) f_1(t) dt \\ & \{ \cos(n\omega_0 t), \sin(n\omega_0 t) \}_{n=-\infty}^{\infty} \text{ 是一个完备正交函数集} \\ & \int_{-\infty}^{\infty} \cos(n\omega_0 t) \sin(m\omega_0 t) dt = 0 \\ & \left. \begin{aligned} \int_{-\infty}^{\infty} \cos(n\omega_0 t) \cos(m\omega_0 t) dt &= \begin{cases} \frac{T}{2}, & n=m \\ 0, & n \neq m \end{cases} \\ \int_{-\infty}^{\infty} \sin(n\omega_0 t) \sin(m\omega_0 t) dt &= \begin{cases} \frac{T}{2}, & n=m \\ 0, & n \neq m \end{cases} \end{aligned} \right\} \end{aligned}$$

2) 级数形式

周期信号 $f(t)$ ，周期为 T ，在满足狄利克雷条件时，可展成：

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

$$a_0 = \frac{\langle f(t), 1 \rangle}{\langle 1, 1 \rangle} = \frac{1}{T} \int_{-\infty}^{\infty} f(t) dt \quad \text{直流分量}$$

$$a_n = \frac{\langle f(t), \cos(n\omega_0 t) \rangle}{\langle \cos(n\omega_0 t), \cos(n\omega_0 t) \rangle} = \frac{2}{T} \int_{-\infty}^{\infty} f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{\langle f(t), \sin(n\omega_0 t) \rangle}{\langle \sin(n\omega_0 t), \sin(n\omega_0 t) \rangle} = \frac{2}{T} \int_{-\infty}^{\infty} f(t) \sin(n\omega_0 t) dt$$

$n=1$ 基波分量 $n>1$ 谐波分量

余弦形式： $f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \phi_n)$

其中： $C_0 = a_0$, $C_n = \sqrt{a_n^2 + b_n^2}$, $\phi_n = \arctan(-\frac{b_n}{a_n})$

$$a_n = C_n \cos \phi_n, b_n = -C_n \sin \phi_n$$

正弦形式： $f(t) = d_0 + \sum_{n=1}^{\infty} d_n \sin(n\omega_0 t + \theta_n)$

$$d_0 = a_0, d_n = \sqrt{a_n^2 + b_n^2}, \theta_n = \arctan(\frac{a_n}{b_n})$$

$$a_n = d_n \sin \theta_n, b_n = d_n \cos \theta_n$$

3) 幅度频率特性和相位频率特性

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \phi_n)$$

$C_n \sim \omega$: 幅度频谱图, 幅度谱

$\phi_n \sim \omega$: 相位频谱图, 相位谱

离散性、谐波性、收敛性

2. 指数形式的傅里叶级数

1) 级数形式

$$\begin{cases} \cos(n\omega_0 t) = \frac{1}{2}(e^{jn\omega_0 t} + e^{-jn\omega_0 t}) \\ \sin(n\omega_0 t) = \frac{1}{2j}(e^{jn\omega_0 t} - e^{-jn\omega_0 t}) \end{cases}$$

$$\Rightarrow f(t) = a_0 + \sum_{n=1}^{\infty} \left[\frac{a_n - jb_n}{2} e^{jn\omega_0 t} + \frac{a_n + jb_n}{2} e^{-jn\omega_0 t} \right]$$

$$\Rightarrow f(t) = a_0 + \sum_{n=1}^{\infty} [F_n e^{jn\omega_0 t} + F_{-n} e^{-jn\omega_0 t}]$$

$$= \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$$

$$F_n = \frac{\langle f(t), e^{jn\omega_0 t} \rangle}{\langle e^{jn\omega_0 t}, e^{jn\omega_0 t} \rangle} = \frac{1}{T} \int_{-\infty}^{\infty} f(t) e^{-jn\omega_0 t} dt$$

$$F_{-n} = F_n^*$$

2) 幅度频率特性和相位频率特性

$$F_n = \frac{a_n - jb_n}{2} = |F_n| e^{j\phi_n}$$

$$|F_n| = \frac{a_n^2 + b_n^2}{4} = \frac{1}{4} C_n^2 \quad F_n = C_n$$

为谐波时基

① 幅频特性： $\phi_n = \arctan(-\frac{b_n}{a_n})$

a_n 是 $f(t)$ 的偶函数 $\Rightarrow \{ |F_n| \}$ 为偶

b_n 是 $f(t)$ 的奇函数 $\Rightarrow \{ \phi_n \}$ 奇

3. 函数对称性与傅里叶级数

1) 偶函数 $f(t) = f(-t)$ 正弦项 + 余弦项

$$a_n = \frac{2}{T} \int_0^{\frac{T}{2}} f(t) \cos(n\omega_0 t) dt = \frac{2}{T} \int_0^{\frac{T}{2}} f(t) \cos(n\omega_0 t) dt$$

$$b_n = 0$$

$$F_n = F_{-n}, \phi_n = -\phi_{-n}, \phi_n = 0, \phi_n > 0$$

2) 奇函数 只有正弦项

$$a_n = 0, a_0 = 0$$

$$b_n = \frac{2}{T} \int_0^{\frac{T}{2}} f(t) \sin(n\omega_0 t) dt \neq 0$$

$$F_n = F_{-n}, \phi_n = -\frac{\pi}{2}, \phi_n > 0$$

3) 奇谐函数 一半周期镜像对称(上下对称) 基波 + 奇次谐波的正弦、余弦项

$$f(t) = -f(t \pm \frac{T}{2}), \omega_0 = \frac{2\pi}{T}$$

$$a_0 = \frac{1}{T} \int_0^T f(t) dt = \frac{1}{T} (\int_0^{\frac{T}{2}} f(t) dt + \int_{\frac{T}{2}}^T f(t) dt)$$

$$= \frac{1}{T} (\int_0^{\frac{T}{2}} f(t) dt + \int_0^{\frac{T}{2}} -f(t) dt) = 0$$

$$a_n = \frac{2}{T} (\int_0^{\frac{T}{2}} f(t) \cos(n\omega_0 t) dt + \int_{\frac{T}{2}}^T f(t) \cos(n\omega_0 t) dt)$$

$$= \frac{2}{T} (\int_0^{\frac{T}{2}} f(t) \cos(n\omega_0 t) dt + \int_0^{\frac{T}{2}} -f(t) \cos(n\omega_0 t + n\pi) dt)$$

$$= \begin{cases} 0, & n \text{ 为偶} \\ \frac{4}{T} \int_0^{\frac{T}{2}} f(t) \cos(n\omega_0 t) dt, & n \text{ 为奇} \end{cases}$$

$$b_n = \begin{cases} 0, & n \text{ 为偶} \\ \frac{4}{T} \int_0^{\frac{T}{2}} f(t) \sin(n\omega_0 t) dt, & n \text{ 为奇} \end{cases}$$

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